

CS 236 Problem Set 2

Francis Dimzon
Joshua Frankie Rayo
Isabelle Tingzon

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10.1a Given

$$u''(t) = f(t)$$

we apply the Fundamental Theorem of Calculus to get,

$$\int_0^t u''(t) = u'(t) = c_1 + \int_0^t f(t)dt$$

where c_1 is determined by the initial condition. We can rewrite this as a definite integral

$$\begin{aligned} u'(t) &= c_1 + \int_0^t f(x)dx \\ &= c_1 + F(t) \end{aligned}$$

where $F(t) = \int_0^t f(x)dx$. Integrating one more time using the Fundamental Theorem of Calculus,

$$\begin{aligned} \int_0^t u'(t) = u(t) &= \int_0^t (c_1 + F(t))dt \\ u(t) &= c_2 + c_1t + \int_0^t F(s)ds \end{aligned}$$

where $F(s) = \int_0^s f(x)dx$

10.1b Show $u(t) = c_1 + c_2t + \int_0^t (t-s)f(s)ds$

From a), $u(t) = c_1 + c_2t + \int_0^t F(s)ds$

It suffice to show that $\int_0^t F(s)ds = \int_0^t (t-s)f(s)ds$

Using integration by parts, $\int u dv = uv - \int v du$

Let $u=F(s)$ and $du=dF(s)$, then $dv = ds$ and $v = s$.

$$\begin{aligned} \int_0^t F(s)ds &= sF(s)|_0^t - \int_0^t s dF(s) = tF(t) - 0F(0) - \int_0^t s dF(s) = tF(t) - \int_0^t s f(s)ds = \\ \int_0^t t f(s)ds - \int_0^t s f(s)ds &= \int_0^t (t-s)f(s)ds. \end{aligned}$$

10.1c Given: $u(0) = 0$

$$\implies 0 = u(0) = c_1 + (c_2)(0) + \int_0^0 F(s)ds$$

$$\implies 0 = c_1$$

Given: $u(1) = 0$

$$\implies 0 = u(1) = c_1 + (c_2)(1) + \int_0^1 F(s)ds$$

$$\implies 0 = 0 + c_2 + \int_0^1 F(s)ds$$

$$\implies 0 = c_2 + \int_0^1 (1-s)f(s)ds$$

$$\begin{aligned}
&\implies 0 = c_2 + \int_0^1 f(s)ds - \int_0^1 sf(s)ds \\
&\implies c_2 = \int_0^1 sf(s)ds - \int_0^1 f(s)ds \\
&\implies c_2 = \int_0^1 (s-1)f(s)ds \\
&\text{Hence, } u(t) = t \int_0^1 (s-1)f(s)ds + \int_0^t (t-s)f(s)ds
\end{aligned}$$

$$\begin{aligned}
\mathbf{10.1d} \quad u(t) &= t \int_0^1 (s-1)f(s)ds + \int_0^t (t-s)f(s)ds \\
&= t \left(\int_0^t (s-1)f(s)ds + \int_t^1 (s-1)f(s)ds \right) + \int_0^t (t-s)f(s)ds \\
&= \int_0^t t(s-1)f(s)ds + \int_t^1 t(s-1)f(s)ds + \int_0^t (t-s)f(s)ds \\
&= \int_0^t tsf(s)ds - \int_0^t tf(s)ds + \int_0^t tf(s)ds - \int_0^t sf(s)ds + \int_t^1 t(s-1)f(s)ds \\
&= \int_0^t tsf(s)ds - \int_0^t sf(s)ds + \int_t^1 t(s-1)f(s)ds \\
&= \int_0^t s(t-1)f(s)ds + \int_t^1 t(s-1)f(s)ds \\
&= \int_0^1 G(t,s)f(s)ds.
\end{aligned}$$

$$\begin{aligned}
\mathbf{10.1e} \quad u(t) &= \int_0^1 G(t,s)f(s)ds \\
&\text{at } f(t) = 1 \\
u(t) &= \int_0^1 G(t,s)(1)ds = \int_0^t s(t-1)ds + \int_t^1 t(s-1)ds = \frac{(t-1)t^2}{2} + t \left(-\frac{t^2-2t}{2} - \frac{1}{2} \right) = \frac{t^2-t}{2}.
\end{aligned}$$

$$\begin{aligned}
\mathbf{10.1f} \quad u(t) &= \int_0^1 G(t,s)f(s)ds \\
&\text{at } f(t) = t \\
u(t) &= \int_0^1 G(t,s)sds = \int_0^t s^2(t-1)ds + \int_t^1 ts(s-1)ds = \frac{t^3-t}{6}.
\end{aligned}$$

$$\begin{aligned}
\mathbf{10.1g} \quad u(t) &= \int_0^1 G(t,s)f(s)ds \\
&\text{at } f(t) = t^2 \\
u(t) &= \int_0^1 G(t,s)s^2ds = \int_0^t s^3(t-1)ds + \int_t^1 ts^2(s-1)ds = \frac{t^4-t}{12}.
\end{aligned}$$

$$\begin{aligned}
\mathbf{10.2a} \quad y_1 &= u, \quad y_2 = u' \\
\mathbf{y}' &= \begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} y_2 \\ y_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \mathbf{A}\mathbf{y} \\
&\text{boundary conditions:} \\
\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} y_1(0) \\ y_2(0) \end{bmatrix} &+ \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} y_1(b) \\ y_2(b) \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}
\end{aligned}$$

10.2b If we have an ODE in the form $\mathbf{y}' = \mathbf{A}\mathbf{y}$, the general solution is given by $y = c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2$ where λ are the eigenvalues and $\mathbf{v}$ are the corresponding eigenvectors. So the general solution of the ODE is

$$\mathbf{y}(t) = c_1 e^{-t} \begin{bmatrix} -1 \\ 1 \end{bmatrix} + c_2 e^t \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

From chapter 10.2, $\mathbf{Y}(t) = [y_1(t) \ y_2(t)]$ where $y_1(t)$ is the specific solution given that $y(0) = [1 \ 0]^T$ and $y_2(t)$ is the specific solution given that $y(0) = [0 \ 1]^T$. Solving for $y_1(t)$ we have

$$\begin{aligned}
1 &= -c_1 + c_2 \\
0 &= c_1 + c_2 \\
y_1(t) &= -\frac{1}{2}e^{-t} \begin{bmatrix} -1 \\ 1 \end{bmatrix} + \frac{1}{2}e^t \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \cosh(t) \\ \sinh(t) \end{bmatrix}
\end{aligned}$$

Solving for $y_2(t)$ we have

$$\begin{aligned} 0 &= -c_1 + c_2 \\ 1 &= c_1 + c_2 \\ y_1(t) &= \frac{1}{2}e^{-t} \begin{bmatrix} -1 \\ 1 \end{bmatrix} + \frac{1}{2}e^t \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \sinh(t) \\ \cosh(t) \end{bmatrix} \end{aligned}$$

$$\text{Hence, } \mathbf{Y}(t) = \begin{bmatrix} \cosh(t) & \sinh(t) \\ \sinh(t) & \cosh(t) \end{bmatrix}$$

10.2c Solutions to the ODE are not stable since $\|\mathbf{Y}(t)\|_\infty$ increases with t exponentially.

$$\begin{aligned} \mathbf{10.2d} \quad \mathbf{Q} &= \mathbf{B}_0 \mathbf{Y}(0) + \mathbf{B}_b \mathbf{Y}(b) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} \cosh(b) & \sinh(b) \\ \sinh(b) & \cosh(b) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \\ &\begin{bmatrix} \cosh(b) & \sinh(b) \\ \sinh(b) & \cosh(b) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \cosh(0) & \sinh(0) \\ \sinh(0) & \cosh(0) \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \cosh(b) & \sinh(b) \\ \sinh(b) & \cosh(b) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \cosh(b) & \sinh(b) \\ \sinh(b) & \cosh(b) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ \cosh(b) & \sinh(b) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ \cosh(b) & \sinh(b) \end{bmatrix}. \end{aligned}$$

$$\begin{aligned} \mathbf{10.2e} \quad \Phi(t) &= \mathbf{Y}(t) \mathbf{Q}^{-1} \\ &= \begin{bmatrix} \cosh(t) & \sinh(t) \\ \sinh(t) & \cosh(t) \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \cosh(b) & \sinh(b) \end{bmatrix}^{-1} \\ &= \begin{bmatrix} \cosh(t) & \sinh(t) \\ \sinh(t) & \cosh(t) \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -\frac{\cosh(b)}{\sinh(b)} & \frac{1}{\sinh(b)} \end{bmatrix} \\ &= \begin{bmatrix} \frac{\sinh(b) \cosh(t) - \cosh(b) \sinh(t)}{\sinh(b)} & \frac{\sinh(t)}{\sinh(b)} \\ \frac{\sinh(b) \sinh(t) - \cosh(b) \cosh(t)}{\sinh(b)} & \frac{\cosh(t)}{\sinh(b)} \end{bmatrix} \\ &= \frac{1}{\sinh(b)} \begin{bmatrix} \sinh(b-t) & \sinh(t) \\ -\cosh(b-t) & \cosh(t) \end{bmatrix} \end{aligned}$$

10.2f We will focus in the ∞ -norm which is the maximum absolute row sum as b goes to infinity.

$$\begin{aligned} \|\mathbf{Q}\|_\infty &= \max(1, \cosh(b) + \sinh(b)) \\ \|\mathbf{Q}^{-1}\|_\infty &= \max(1, \coth(b) + \operatorname{csch}(b)) \\ \|\Phi\|_\infty &= \frac{1}{e^t} \end{aligned}$$

As b increases to infinity, the first norm also increases without bound while the second norm goes to 1. So, $\operatorname{cond}(\mathbf{Q})$ increases without bound as b increases. The unique solution to the BVP is given by

$$\begin{aligned}\mathbf{y}(t) &= \Phi(t) \cdot [0 \quad \beta]^T \\ \|\mathbf{y}(t)\|_\infty &= \|\Phi(t)\|_\infty \cdot \beta \\ \|\mathbf{y}(t)\|_\infty &= \frac{\beta}{e^t}\end{aligned}$$

The solution is bounded by the function given above, which does not blow up when $t \in [0, \infty]$.

10.3a Using Euler's method from $t=a$ to $t=b$

$$\begin{aligned}u(b) &= u(a) + (b-a)slope_{initial} \\ \implies \beta &= \alpha + (b-a)slope_{initial}\end{aligned}$$

10.3b $slope_{initial} = \frac{\beta-\alpha}{b-a}$

10.4a $y_1 = u, \quad y_2 = u'$

$$\mathbf{y}' = \begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} y_2 \\ -4 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} + \begin{bmatrix} 0 \\ -4 \end{bmatrix} = \mathbf{A}\mathbf{y} + \mathbf{b}$$

1) Trapezoid Method $y_{k+1} = y_k + h(y_k' + y_{k+1}')/2$

$$\begin{bmatrix} 1 \\ s_1 \end{bmatrix} = \begin{bmatrix} 1 \\ s_0 \end{bmatrix} + \frac{1}{2} \left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ s_0 \end{bmatrix} + \begin{bmatrix} 0 \\ -4 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ s_1 \end{bmatrix} + \begin{bmatrix} 0 \\ -4 \end{bmatrix} \right)$$

2. From 1), $s_0 + s_1 = 0$ and $s_1 = s_0 - 4$

$$\implies s_0 = 2 \text{ and } s_1 = -2.$$

$$3. \begin{bmatrix} y_1(0.5) \\ y_2(0.5) \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \frac{0.5}{2} \left(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ -4 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} y_1(0.5) \\ y_2(0.5) \end{bmatrix} + \begin{bmatrix} 0 \\ -4 \end{bmatrix} \right)$$

$$\implies y_1(0.5) = 1.5, \quad y_2(0.5) = 0$$

$$\implies u(0.5) \approx 1.5$$

10.4b $u'' = -4$

$$\implies \frac{y_0 - 2y_1 + y_2}{h^2} = -4$$

$$\implies \frac{1 - 2y_1 + 1}{0.5^2} = -4$$

$$\implies y_1 = 1.5$$

$$\implies u(0.5) = 1.5$$

10.4c $u(t) \approx x_0 + x_1 t + x_2 t^2$

$$u(0) = 1 = x_0 + (x_1)(0) + (x_2)(0^2)$$

$$\implies x_0 = 1$$

$$u(1) = 1 = x_0 + (x_1)(1) + (x_2)(1^2)$$

$$\implies x_1 = -x_2$$

$$u''(t) = 2x_2 = -4$$

$$\implies x_2 = -2$$

$$\implies x_1 = 2$$

$$u(t) \approx 1 + 2t - 2t^2$$

$$u(0.5) \approx 1 + 2(0.5) - 2(0.5)^2 = 1.5$$

11.1a $\int_a^b g(t) dt$

Let $y(t) = \int_a^t g(x) dx$. Solving $\int_a^b g(t) dt$ is equivalent to finding the value of $y(b)$.

1. Solving the problem becomes solving the ODE subproblem $y'(t) = g(t)$.

2. The initial condition is that $y(a) = 0$.

3. No special properties unless $g(t)$ is known.

11.1b $y''(t) = y^2 + t, \quad 0 \leq t \leq 1 \quad y(0) = 0, \quad y(1) = 1$

Converting to first order ODE system, we have

$$u_1 = y, \quad u_2 = y'$$

$$\mathbf{u}' = \begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} u_2 \\ u_1^2 + t \end{bmatrix}$$

1. This is solvable by the ODE solver. The BVP can be solved using shooting method.
2. $u_1(0) = 0$. $u_2(0)$ is obtained by several guesses until $u_1(1) = 1$.
3. No special properties

11.1c $u_t = cu_{xx}, \quad 0 \leq x \leq 1, \quad t \geq 0$

$$u(0, x) = g(x), \quad 0 \leq x \leq 1$$

$$u(t, 0) = 0, \quad u(t, 1) = 0, \quad t \geq 0$$

1.

$$y_i'(t) = \frac{c}{(\Delta x)^2} (y_{i+1} - 2y_i + y_{i-1})$$

which can be solved using method of lines

2. Initial conditions: $y_i(0) = g(x_i), \quad i = 1, \dots, n$
3. The ODE problem is stiff and the solver must deal with stiff problems

11.2 Substituting the centered finite difference approximations in the PDE,

$$\frac{u_{x+\Delta x, y} - 2u_{x, y} + u_{x-\Delta x, y}}{(\Delta x)^2} + \frac{u_{x, y+\Delta y} - 2u_{x, y} + u_{x, y-\Delta y}}{(\Delta y)^2} \approx x + y.$$

$$\implies \frac{1 - 2u_{x, y} + 0}{(0.5)^2} + \frac{1 - 2u_{x, y} + 0}{(0.5)^2} \approx 0.5 + 0.5$$

$$\implies u_{x, y} \approx 0.4375$$

11.4 Consider the Laplace equation $\nabla^2 u = 0$ with $u = 0$ at the boundary to be solve using fully discrete methods, using $k = 5$ interior mesh points for each dimension.

For the 1-dimensional case, we have $u_{xx} = 0$. Using centered finite difference of second derivatives, the resulting matrix will be tridiagonal:

$$\mathbf{A} = \begin{bmatrix} -2 & 1 & 0 & \cdots & 0 \\ 1 & -2 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & 1 & -2 & 1 \\ 0 & \cdots & 0 & 1 & -2 \end{bmatrix}$$

Numerical values of nonzero entries include 1, -2 and 1.

For the 2-dimensional case, we have $u_{xx} + u_{yy} = 0$. The resulting matrix will be also tridiagonal:

$$X_1 = \begin{bmatrix} -2 & 1 & 0 & \cdots & 0 \\ 1 & -2 & 1 & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & 1 & -2 & 1 \\ 0 & \cdots & 0 & 1 & -2 \end{bmatrix}$$

$$X_n = \begin{bmatrix} n/(n-1) X_{n-1} & I_{k^{(n-1)}} & 0 & \cdots & 0 \\ I_{k^{(n-1)}} & n/(n-1) X_{n-1} & I_{k^{(n-1)}} & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & I_{k^{(n-1)}} & n/(n-1) X_{n-1} & I_{k^{(n-1)}} \\ 0 & \cdots & 0 & I_{k^{(n-1)}} & n/(n-1) X_{n-1} \end{bmatrix}$$

Numerical values of nonzero entries include 1, -2n, and 1.

11.6 Prove that the Jacobi iterative method for solving the linear system $Ax = b$ converges if the matrix A is diagonally dominant by rows.

Proof. By definition, A is diagonally dominant if $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$. In other words, we can define r

$$r = \sum_{j \neq i} \frac{|a_{ij}|}{|a_{ii}|} < 1$$

A linear iteration method converges for any starting iterate and any right hand side provided that matrix norm $\|G\| < 1$. Recall that Jacobi iteration scheme is given by

$$x^{k+1} = -D^{-1}(L + U)x^{(k)} + D^{-1}b$$

where $A = D + L + U$.

Therefore, for convergence we need $\|G\| = \|-D^{-1}(L + U)x^{(k)}\| < 1$. The ∞ -norm of a matrix M is given by

$$\|M\|_{\infty} = \max_{1 \leq i \leq n} \sum_{j=1}^n |m_{ij}|$$

If A is diagonally dominant by rows, then the following must be true

$$\|-D^{-1}(L + U)x^{(k)}\|_{\infty} = \max \sum_{j \neq i} \frac{|a_{ij}|}{|a_{ii}|} \leq r < 1$$

Therefore, the Jacobi iterative method for solving the linear systems converges if matrix A is diagonally dominant by rows.

11.7 Let B be any $n \times n$ complex matrix with zero diagonal entries which we write in the form $B=L+U$, where L and U are, respectively, strictly lower and strictly upper triangular matrices. If

$$L_{\omega} := (I - \omega L)^{-1}(\omega U + (1 - \omega)I),$$

then for any real or complex number ω ,

$$\rho(L_{\omega}) \geq |\omega - 1|,$$

with equality holding only if all eigenvalues of L_{ω} are of modulus $|\omega - 1|$.

Proof. Let $\phi(\lambda) := \det(\lambda I - L_\omega)$ be the characteristic polynomial of L_ω . Since L is strictly lower triangular, then $(I - \omega L)$ is nonsingular, and as $\det(I - \omega L) = 1$, then

$$\phi(\lambda) = \det(I - \omega L) \cdot \det(\lambda I - L_\omega) = \det\{(\lambda + \omega - 1)I - \omega \lambda L - \omega U\}.$$

The constant term μ of $\phi(\lambda)$, the product of the negatives of the eigenvalues of L_ω , is obtained by setting $\lambda = 0$ in this expression. Thus,

$$\mu = \prod_{i=1}^n (-\lambda_i(\omega)) = \det\{(\omega - 1)I - \omega U\} = (\omega - 1)^n.$$

Therefore,

$$\rho(L_\omega) = \max_{1 \leq i \leq n} |\lambda_i(\omega)| \geq |\omega - 1|,$$

with equality possible only if all eigenvalues of L_ω are modulus $|\omega - 1|$, which completes the proof. If we wish to determine the *real* value of ω which minimizes $\rho(L_\omega)$, then the above result of Kahan shows that we need only consider the open interval $0 < \omega < 2$ for values of ω such that L_ω is convergent.

11.8